

Normalization Report (Up to BCNF)

Bank Management System for Gebar Commercial Bank

Database Management Systems Project

Addis Ababa Science and Technology University

1. Unnormalized Form (UNF)

All data is initially stored in a single table with repeating groups and redundancy.

CustomerID	CustomerName	Accounts	Loans	Transactions
CUST-10045	Selam Tesfaye	ACC-1, ACC-2	LN-1	TRX-1, TRX-2

2. Functional Dependencies

The following functional dependencies were identified:

Dependency
CustomerID → FirstName, LastName, Phone
AccountNumber → CustomerID, AccountType, Balance
LoanID → CustomerID, LoanAmount, LoanType
TransactionID → AccountNumber, Amount, TransactionType

3. First Normal Form (1NF)

Eliminate repeating groups and ensure atomic values.

CustomerID	FirstName	LastName	AccountNumber	LoanID	TransactionID
CUST-10045	Selam	Tesfaye	ACC-1	LN-1	TRX-1

4. Second Normal Form (2NF)

Remove partial dependencies by splitting into separate tables.

CustomerID	FirstName	LastName
CUST-10045	Selam	Tesfaye

AccountNumber	CustomerID	AccountType
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ACC-1	CUST-10045	Savings
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5. Third Normal Form (3NF)

Remove transitive dependencies.

LoanID	CustomerID	LoanType	LoanAmount
LN-1	CUST-10045	Education Loan	85000

TransactionID	AccountNumber	Amount	TransactionType
TRX-1	ACC-1	25000	Deposit

6. Boyce-Codd Normal Form (BCNF)

All determinants are candidate keys. The schema is fully normalized.

Final Relations:

Customer(CustomerID, FirstName, LastName, Phone)

Account(AccountNumber, CustomerID, AccountType, Balance)

Loan(LoanID, CustomerID, LoanType, LoanAmount)

Transaction(TransactionID, AccountNumber, Amount, TransactionType)